

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

28th November 2020

**Subject CM2A – Financial Engineering and Loss
Reserving (Paper A)**

Time allowed: 3 Hours 30 Minutes (09.30 – 13.00 Hours)

Total Marks: 100

Q. 1) You have two choices of assets, A and B having the following expected returns:

State	Return A	Return B	Probability
Good	20%	2%	0.3
Bad	4%	10%	0.2
Average	10%	6%	0.5

i) The expected returns for Asset A and Asset B are:

- a. 10.2% and 6.4%
- b. 11.8% and 5.2%
- c. 11.8% and 5.6%
- d. 12.2% and 5.2%

(1)

ii) The S.D of returns for Asset A and Asset B are:

- a. 5.83% and 2.80%
- b. 5.55% and 2.80%
- c. 5.55% and 2.81%
- d. 5.83% and 2.81%

(2)

iii) Calculate the correlation coefficient between the returns on A and B.

- a. 0.9792
- b. - 0.9855
- c. 0.9750
- d. - 0.9818

(2)

iv) Comment on your answer in part (iii).

(1)

[6]

Q. 2) Explain how Behavioural Finance arguments might be used by an equity investment manager to explain why they hold a persistently underperforming stock in a portfolio rather than selling out of the position and re-investing in an alternative stock.

[5]

Q. 3) As the Investment Manager of a hedge fund in Actuarial, you are considering an opportunity to invest in an exotic venture. The modelling team has suggested that the relative returns (x) from the investment exhibit the following probability density function (PDF):

$$f(x) = \min\{x + 1, e^{-cx}\} \text{ for } x \text{ in } [-1, \infty)$$

where, c is a positive constant.

Please note that the lowest possible value for relative return from any investment is -1 (or -100%, corresponding to complete loss of invested amount), while there is no highest possible value (in theory, the returns can be arbitrarily large).

i) Find the interval for which $f(x)$ equals $(x+1)$.

- a. $[-1, 0]$
- b. $[0, \infty)$
- c. $[-1, \infty)$
- d. $[-1, 1]$

(1)

ii) Find the value of c for which the given function is a valid PDF.

- a. 0.5
- b. 1
- c. 2
- d. 4

(2)

iii) Compute the expected return from this exotic investment.

- a. 0
- b. $1/4$
- c. $1/6$
- d. $1/12$

(3)

iv) Evaluate $E[X^2]$.

- a. $1/12$
- b. $1/4$
- c. $1/3$
- d. $5/12$

(3)

v) Compute the variance of X .

(1)

vi) Compute the Value at Risk (VaR) at 90th percentiles assuming a \$1m investment.

(3)

vii) Compute the shortfall probability and the expected shortfall (again assuming a \$1m investment) with a benchmark return level of -50%.

What does the use of a shortfall risk measure suggest about the investor's utility function?

(4)

Note: Sub-parts i to iv are multiple choice questions, where it suffices to choose the correct option.

[17]

Q. 4) The stock exchange in the country of Actuarialia is a flourishing one with many stocks and derivatives traded. The Actuarial Stock Exchange maintains its benchmark index called ACTEX, with futures and European options traded for this index as well. Following is the snapshot of option prices expiring 6 months from now with some entries missing:

Strike (\$)	Call Price	Put Price
10000	1990.88	
11000	1342.92	625.11
12000	856.82	1119.21

- i) Fill in the missing value in the above table. (4)
 - ii) If a future contract on ACTEX is entered into today to be settled 6 months from now, what should be the fair price as set today? (1)
 - iii) If, instead of the index, the future contract was on a particular stock, called Resilience, currently trading at \$200 and a \$6 dividend payable 2 months from now, what should be the fair price as set today for the 6-month future contract? (2)
 - iv) State the assumptions underlying the binomial model. (2)
 - v) Derive from first principles the value of a one-month put option on the stock at strike price \$205 under the one-period binomial model. Assume that the stock can move up to \$210 with a 60% probability and move down to \$195 with a 40% probability. (5)
 - vi) If a forward contract on a non-dividend paying stock called Growth Industries (Resilience currently trading at S), expiring after time $(T-t)$ from today was entered into in the past for a strike price of K , write the expression for the current value of a long position of the forward contract. (1)
 - vii) In order to show that this expression above satisfies the Black Scholes PDE, we need to compute delta (Δ), gamma (Γ) and theta (Θ). Write expressions for all of them. (2)
 - viii) Also show that the expression satisfies the boundary condition expected of a forward contract. (1)
- [18]**

Q. 5) A company has issued a 10 year debt with a nominal face value of 10 cr. The overall asset value of the company is 30 cr.

Assuming we use the Merton model to value the debt issued by the company.

- i) Say the total asset value of the company after 10 years is denoted by $F(10)$, payoff to the bond holder would be as follows:
 - a. $\text{Max}[10, F(10)]$
 - b. $\text{Min}[10, F(10)]$
 - c. $10 - \text{Min}[30 - F(10), 0]$
 - d. $10 - \text{Max}[0, F(10) - 10]$ (2)
- ii) The pay-off for the equity holders of the company can be regarded as having:
 - a. European call option with maturity of 10 years and strike price of 10 cr.
 - b. European call option with maturity of 10 years and strike price of 20 cr.
 - c. European put option with maturity of 10 years and strike price of 20 cr.
 - d. European put option with maturity of 10 years and strike price of 10 cr. (2)
- iii) Calculate the fair price of the debt issued assuming the annualized volatility of the company's assets over the 10 year period is 30% and risk free force of interest is 6% per annum. (5)

[9]

- Q. 6)** An investor has decided to model the share price of ABC using the continuous-time lognormal model. Using historical data, the investor has estimated the annual drift and volatility parameters to be 8% and 25%.

The current share price for ABC is Rs. 1200.

- i) In developing the lognormal model, the investor assumes that the returns on stock prices are mean reverting (True or False) (1)

- ii) The mean and variance of ABC share price in exact 2 years will be:

- a. $1384, 505^2$
- b. $1341, 341^2$
- c. $1499, 547^2$
- d. $1452, 369^2$

(3)
[4]

- Q. 7)** In the Vasicek model, the spot rate of interest is governed by the stochastic differential equation (SDE), $dr_t = 0.8(0.04 - r_t) dt + 0.1 dB_t$

where B_t is a standard Brownian motion.

- i) By considering the function $f(r_t, t) = r_t e^{0.8t}$, or otherwise, solve for this SDE. (5)

- ii) Determine the probability function of r_t and the limiting distribution for large t . (4)

- iii) For $s < t$, show that the conditional expectation $E(r_t | F_s)$ where $\{F_s : s \geq 0\}$ is the filtration generated by the Brownian motion B_s is as follows: (5)

$$E[r_t | F_s] = 0.04(1 - e^{0.8(s-t)}) + e^{0.8(s-t)} r_s$$

[14]

- Q. 8)** Answer the following:

- i) Describe informational efficiency in the context of the Efficient Markets Hypothesis. (3)

- ii) List examples of effects that have been claimed to exist in stock markets that might be considered examples of informational inefficiency. (3)

[6]

- Q. 9)** A salesperson, who is risk-averse and non-satiated, has been offered a one year contract with incentive scheme. His basic salary will be Rs. 70,000. In addition, he will be paid an extra Rs. 30,000 if the company sales exceed a certain target. The probability of the company making enough sales is 0.60.

- i) His utility function is of the form $U(w) = w - 0.3w^2$ (w is expressed as a proportion of Rs 100,000). Derive the remuneration range over which $U(w)$ gives an appropriate representation of his individual preferences. (2)

- ii) Calculate the expected total remuneration and the expected utility offered by the job. (2)

- iii) If he is to be offered the alternative of a fixed salary, calculate the minimum he should accept. (3)
[7]

Q. 10) Claims occur on a portfolio of insurance policies according to a Poisson process at a rate λ . The mean claim amount is μ . The insurance company calculates premiums using a loading of θ and has an initial surplus of U ($< \mu$). Premiums are received continuously.

- i) Explain how the parameters λ , μ , θ and U affect $\psi(U)$, the probability of ultimate ruin. (4)
- ii) The probability that ruin occurs at the first claim is:
- $e^{-1/(1+\theta)*(1-U/\mu)}$
 - $1-e^{-1/(1+\theta)*(1-U/\mu)}$
 - $e^{-\mu/(1+\theta)*(1-U/\mu)}$
 - $1-e^{-\theta/(1+\theta)*(1-U/\mu)}$

(2)

[6]

Q.11 The table below sets out incremental claims data for a portfolio of insurance policies.

Accident year	Development year		
	0	1	2
2016	1574	635	230
2017	1820	942	
2018	2012		

Past and projected future inflation is given by the following index (measured to the mid-point of the relevant year).

Year	Index
2016	100
2017	105
2018	112
2019	115
2020	118

Estimate the outstanding claims using the inflation adjusted chain ladder technique.

[8]
